

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination -2023

Course: B. Tech.

Branch: Civil Engineering

Semester: V

Subject Code & Name: BTCVPE506G_Materials, Testing & Evaluation

Max Marks: 60

Date:22/08/2023

Duration: 3 Hrs.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Enlist various types of materials used in Civil Engineering and Enlist their basic properties.	CO1	6
B) Describe stress-strain behavior in detail.	CO1	6
C) Explain elastic properties of materials.	CO1	6
Q.2 Solve Any Two of the following.		12
A) Illustrate mechanical properties of steel.	CO1	6
B) Discuss uses of wood in Civil Engineering.	CO1	6
C) Illustrate strength of cement and concrete.	CO2	6
Q. 3 Solve Any Two of the following.		12
A) Explain in brief Steel-concrete composite bridge decks.	CO2	6
B) Describe FRP structurally insulated panels in detail.	CO3	6
C) Compare building materials in the context of environmental impact.	CO3	6
Q.4 Solve Any Two of the following.		12
A) Illustrate self-healing concrete.	CO1	6
B) Describe the use of bamboo in construction.	CO3	6
C) Explain zero-cement concrete.	CO2	6
Q. 5 Solve Any Two of the following.		12
A) Explain necessity of material testing.	CO3	6
B) Explain CTM in detail and its applications.	CO2	6
C) Discuss Ultrasonic test in detail.	CO2	6

***** End *****